Linear-size ℓ_1 Sparsifiers

Victor Reis

Joint work with Thomas Rothvoss

Simons Institute Workshop: Beyond Graph Laplacians

November 2, 2026

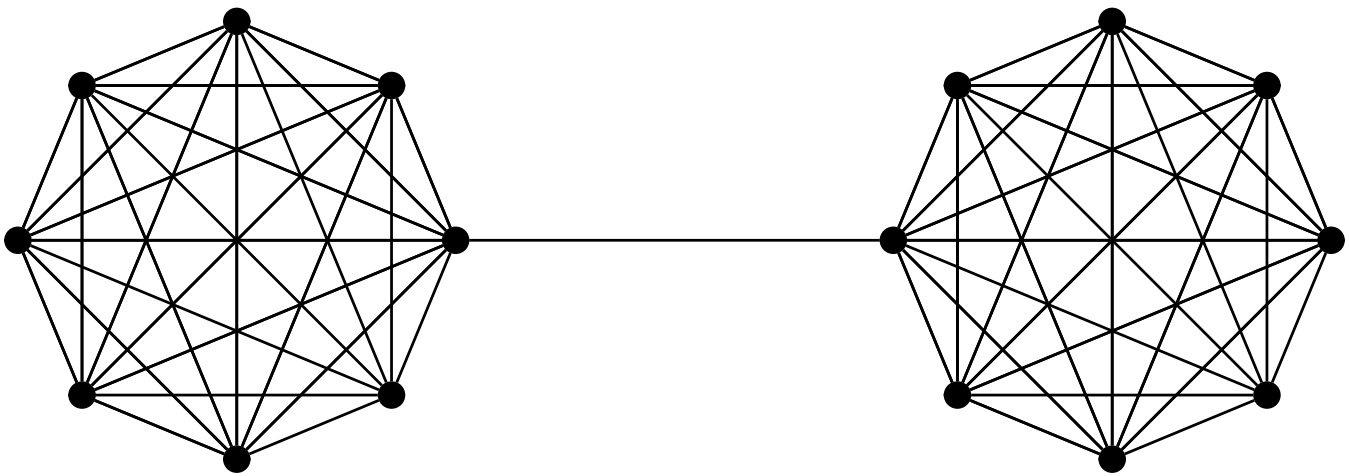




Graph Sparsification

Theorem (Batson-Spielman-Srivastava '08)

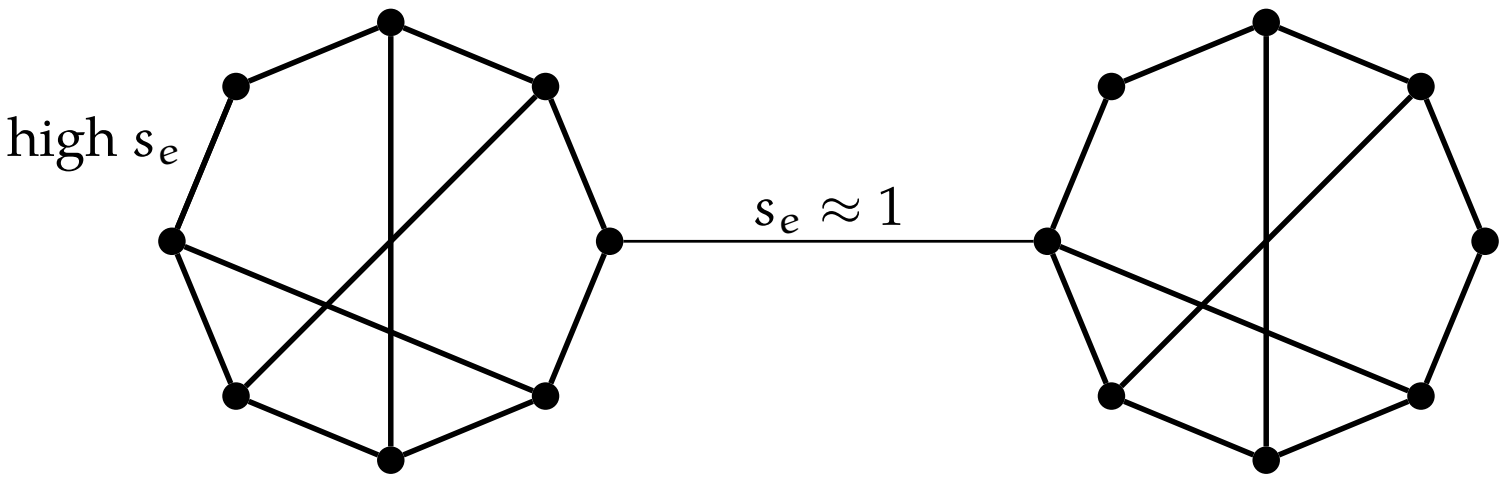
For any graph $G = (V, E)$ one can find weights $s(e) \geqslant 0$ in poly-time with $|\mathrm{supp}(s)| \leqslant O(n/\varepsilon^2)$ so that $|\delta(U)| = (1 \pm \varepsilon) \cdot |s(\delta(U))|$ for every $U \subseteq V$.



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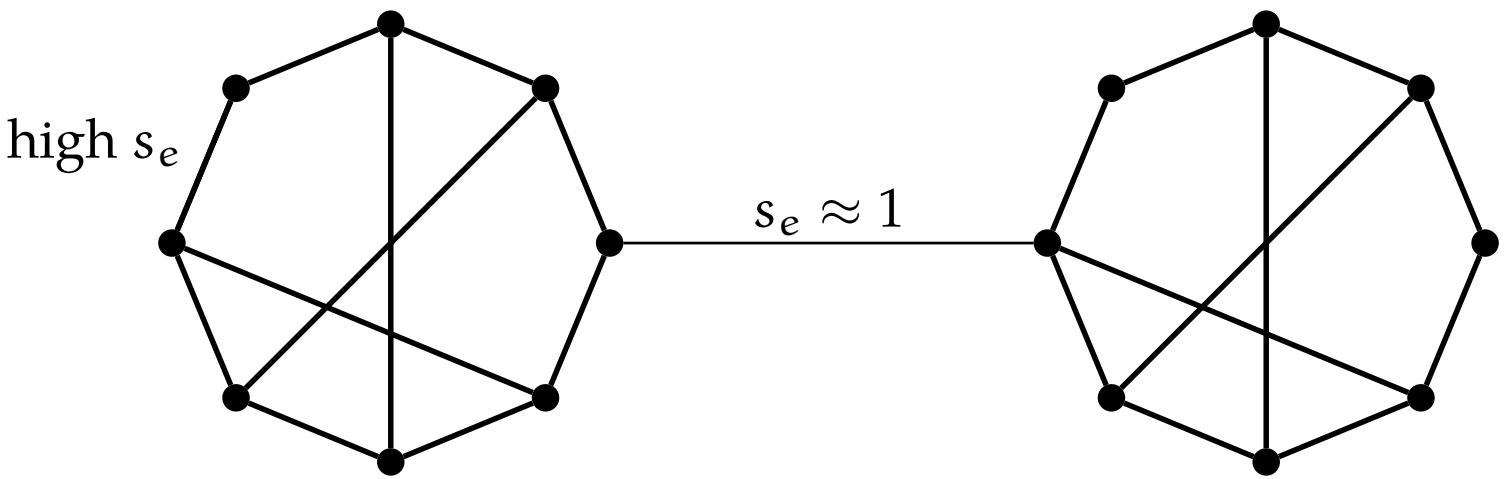
Graph Sparsification

Theorem (Batson-Spielman-Srivastava '08)

For any $A \in \mathbb{R}^{m \times n}$ there is a diagonal matrix $D \in \mathbb{R}^{m \times m}_{\geqslant 0}$ with support size $O(\frac{n}{\varepsilon^2})$ so that

$$(1-\varepsilon)\|Ax\|_2 \leqslant \|DAx\|_2 \leqslant (1+\varepsilon)\|Ax\|_2 \quad \forall x \in \mathbb{R}^n.$$

Matrix D can be found in polytime.



Main result

Theorem (R., Rothvoss'26)

For any matrix $A \in \mathbb{R}^{m \times n}$ and $\varepsilon \in (0, \frac{1}{2}]$, there is a diagonal matrix $D \in \mathbb{R}^{m \times m}_{\geqslant 0}$ with $O(\frac{n}{\varepsilon^2})$ nonzero entries so that

$$(1-\varepsilon)\|Ax\|_1 \leqslant \|DAx\|_1 \leqslant (1+\varepsilon)\|Ax\|_1 \quad \forall x \in \mathbb{R}^n.$$

- Previously known: $O(\frac{n}{\varepsilon^2} \log n)$ [Talagrand '90].
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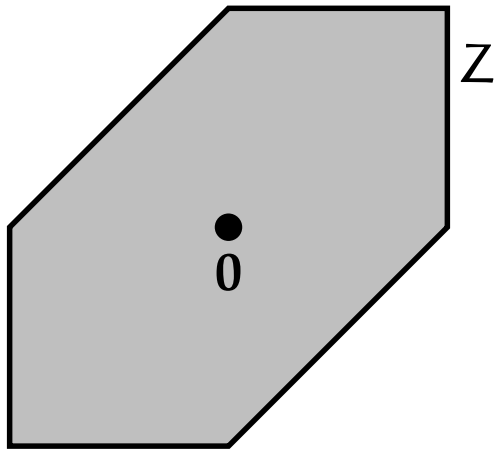
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- No polytime algorithm (given A, B, checking whether $\|Ax\|_1 \leqslant \|Bx\|_1 \ \forall x \in \mathbb{R}^n$ is **coNP**-complete)

Zonotopes

Definition

A **zonotope** $Z \subseteq \mathbb{R}^n$ is the projection of a cube.

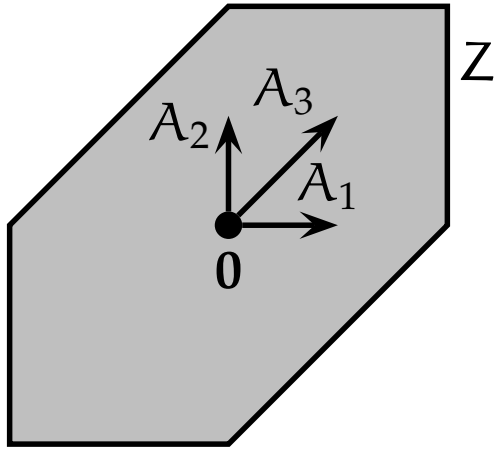


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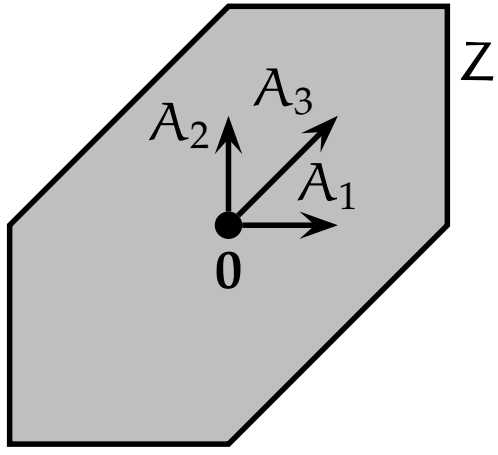
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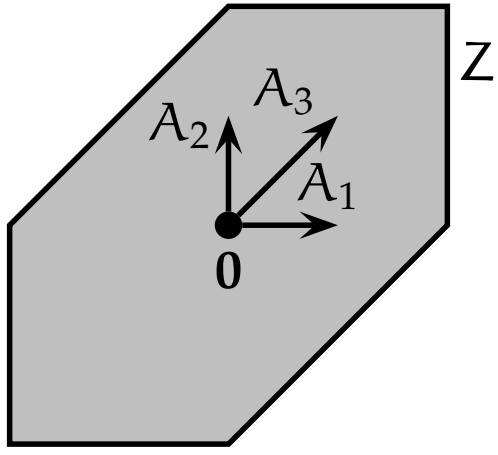
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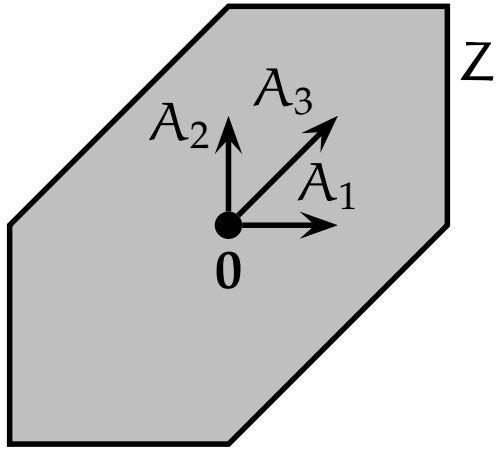
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- m is the **number of segments** of Z
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- **Support function** of zonotope Z:

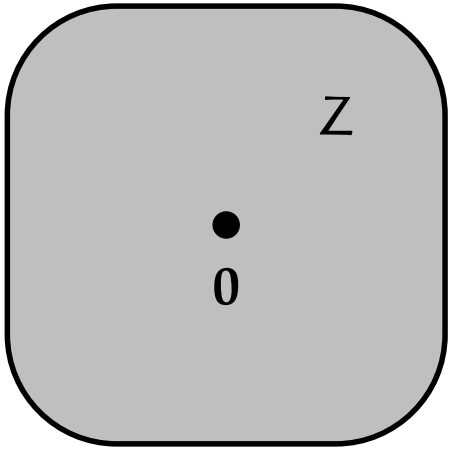
$$h_Z(x) = \max_{y \in Z} \langle y, x \rangle = \|Ax\|_1$$

Reducing number of segments

Theorem (R., Rothvoss'26)

For any zonotope $Z \subseteq \mathbb{R}^n$ and $\varepsilon \in (0, \frac{1}{2}]$, there is a zonotope $Z' \subseteq \mathbb{R}^n$ with $O(\frac{n}{\varepsilon^2})$ segments so that

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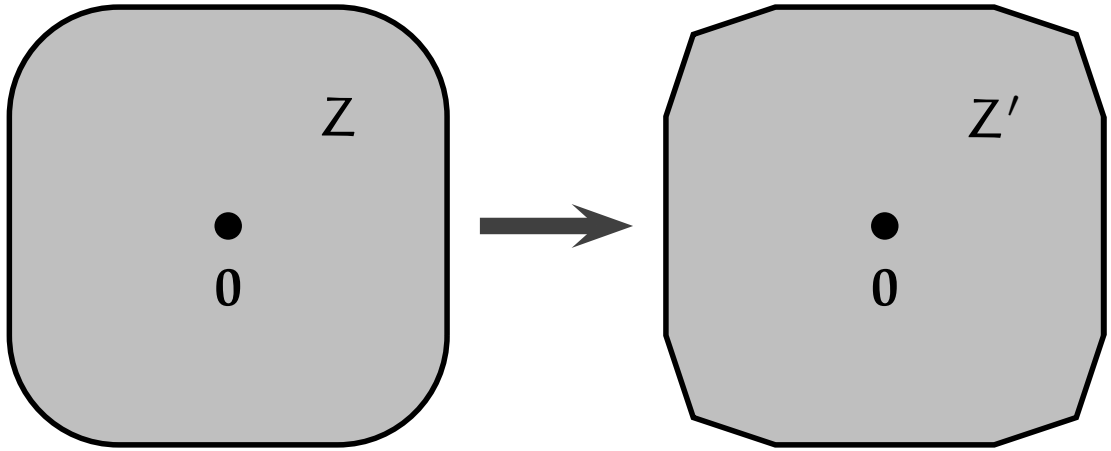


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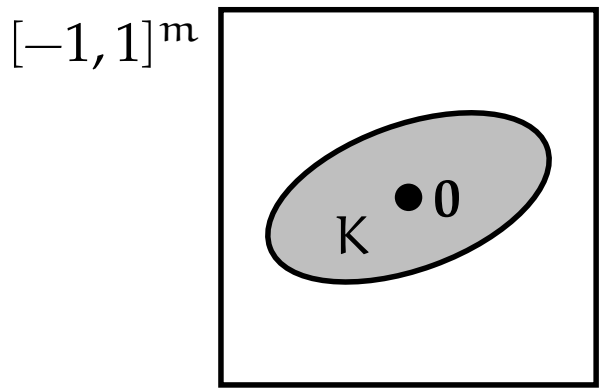
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Discrepancy theory

Theorem

For symmetric convex body $K \subseteq [-1,1]^m$ with $\mathrm{Vol}_m(K) \geqslant c^m$, there is an $x \in s(c)K \cap [-1,1]^m$ with $|\{j \in [m] : |x_j| = 1\}| \geqslant \frac{m}{2}$.

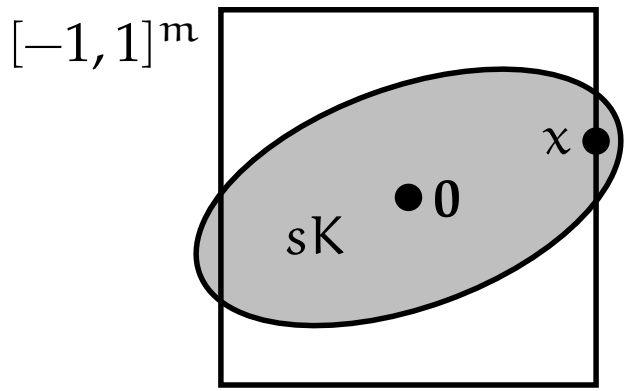


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The overall strategy

(1) Consider

$$K = \left\{ x \in [-1,1]^m : \sum_{i=1}^{m} (1+x_i) \cdot |\langle a_i, y \rangle| \leqslant (1+\varepsilon) \sum_{i=1}^{m} |\langle a_i, y \rangle| \, \forall \, y \in \mathbb{R}^n \right\}$$

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$$K \cap -K = \left\{ x \in [-1,1]^m : \left| \sum_{i=1}^{m} x_i \cdot |\langle a_i, y \rangle| \right| \leqslant \varepsilon \sum_{i=1}^{m} |\langle a_i, y \rangle| \, \forall \, y \in \mathbb{R}^n \right\}$$

also has $\mathrm{Vol}_m(K \cap -K) \geqslant \tilde{c}^m$.

(4) Then there is $x \in s(\tilde{c}) \cdot (K \cap -K)$ with $\frac{m}{4}$ coordinates $x_i = -1$.

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GAP

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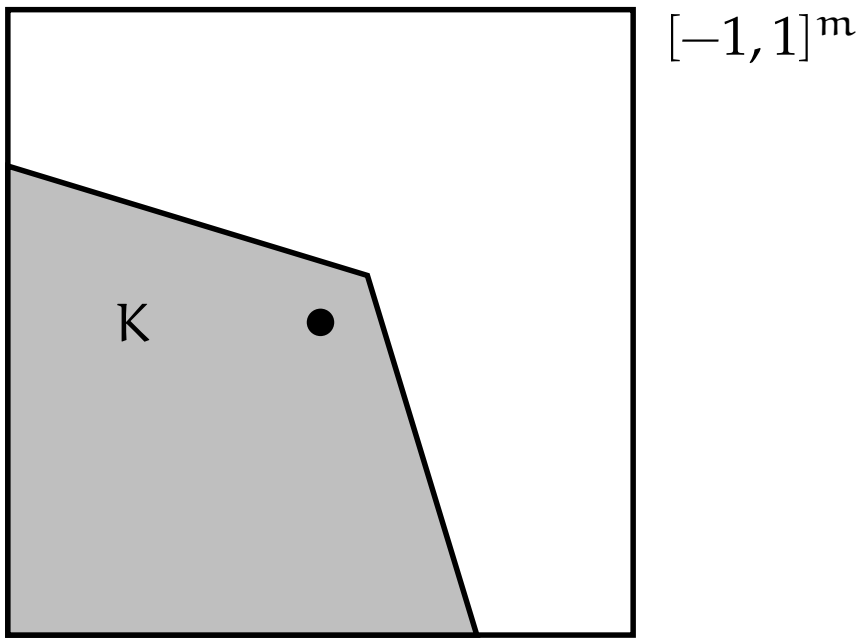
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Size of the symmetrizer (1)

Theorem

Let $K \subseteq [-1,1]^m$ be convex with $\mathrm{Vol}_S(K_S) \geqslant 2^{-\varepsilon^2 m} \cdot 2^{|S|}$ for all $S \subseteq [m]$ and $[-\varepsilon,\varepsilon]^m \subseteq K$. Then

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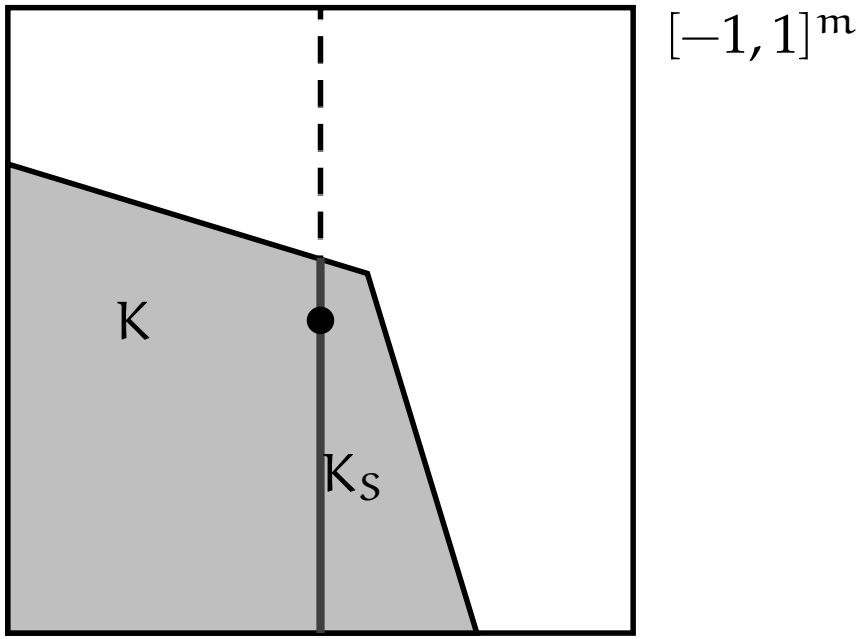


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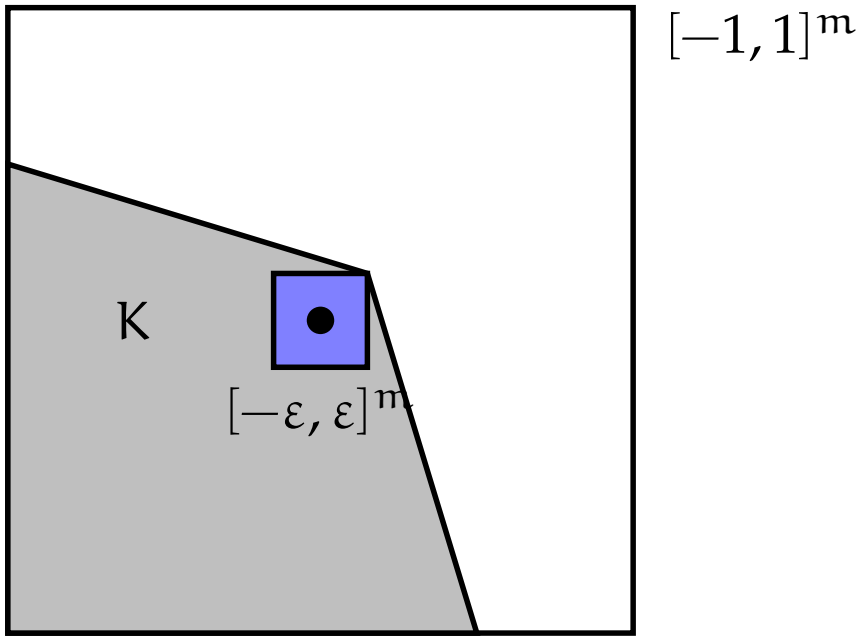


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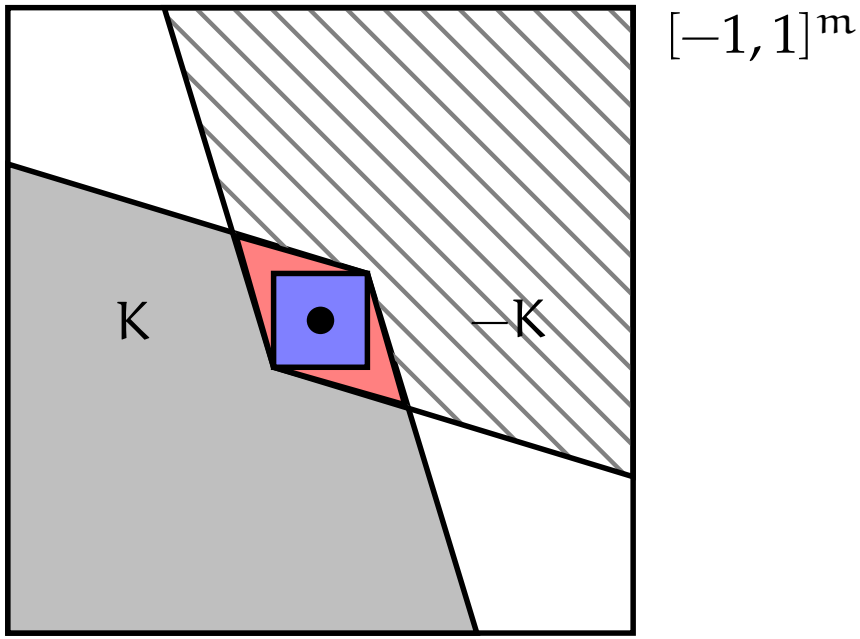


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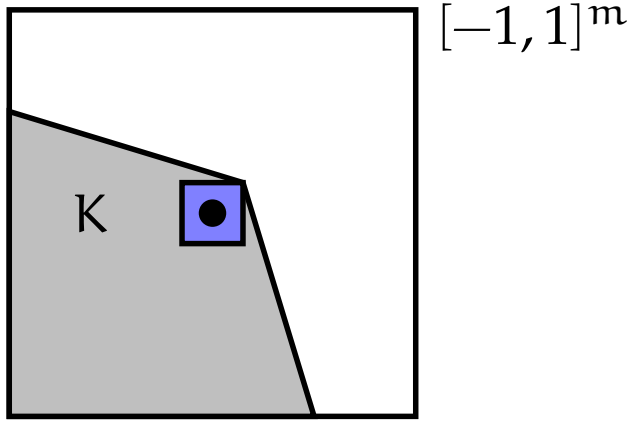
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Theorem (Milman, Pajor)

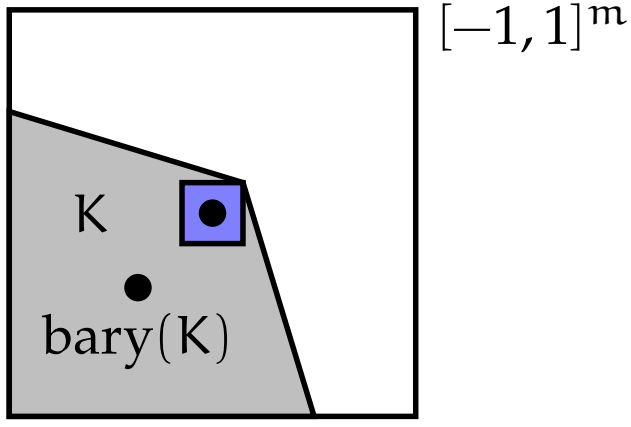
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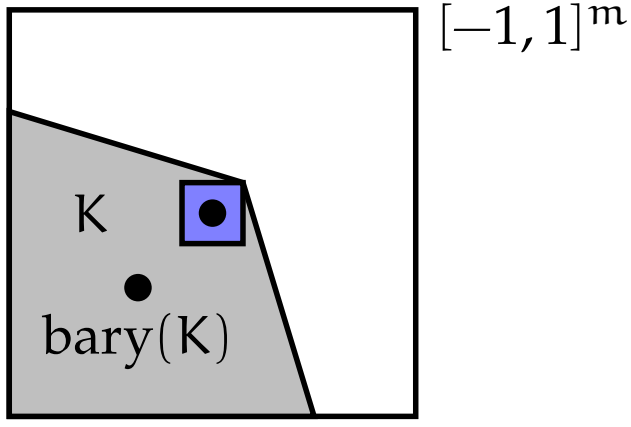
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$$f_K(z) := \mathrm{Vol}_m((K - z) \cap -(K - z))$$

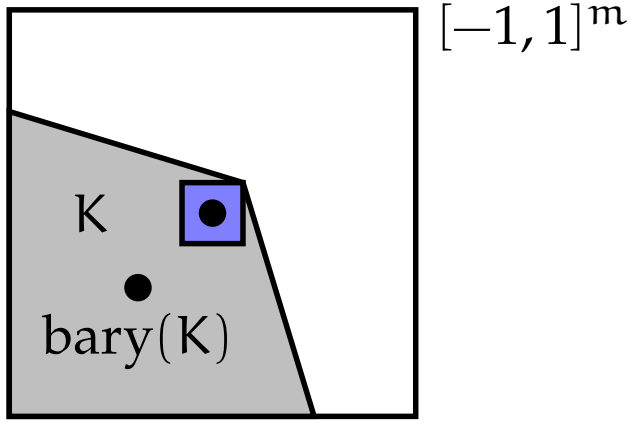


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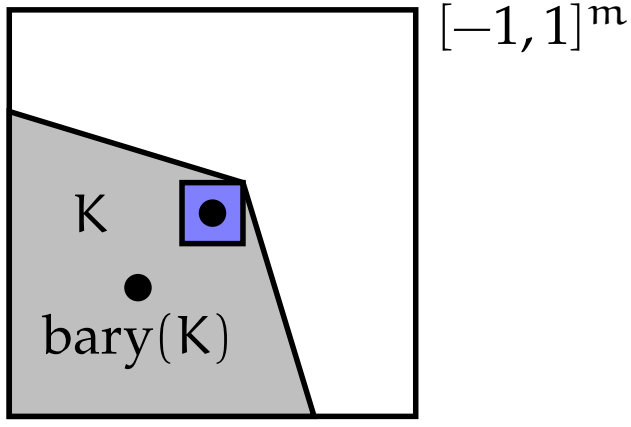


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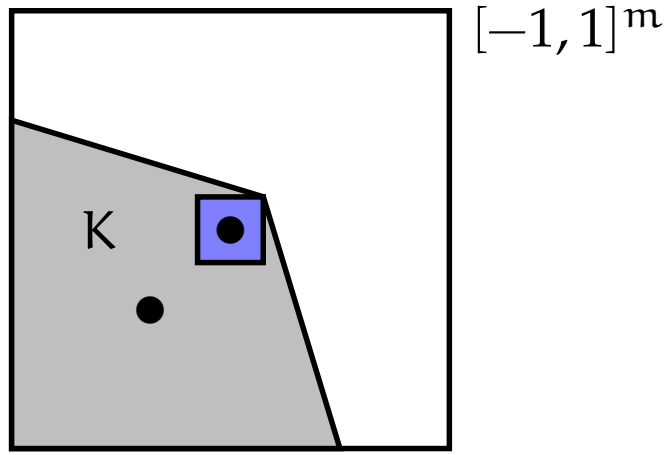
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$\left[\ \textbf{Proof.}\ \frac{1}{2}((K - z) \cap -(K - z)) \subseteq K \cap -K\ \right]$.



Size of the symmetrizer (3)

Lemma

There is a $z \in [-\varepsilon, \varepsilon]^m$ so that $f_K(z) \geqslant 2^{-4m}$.

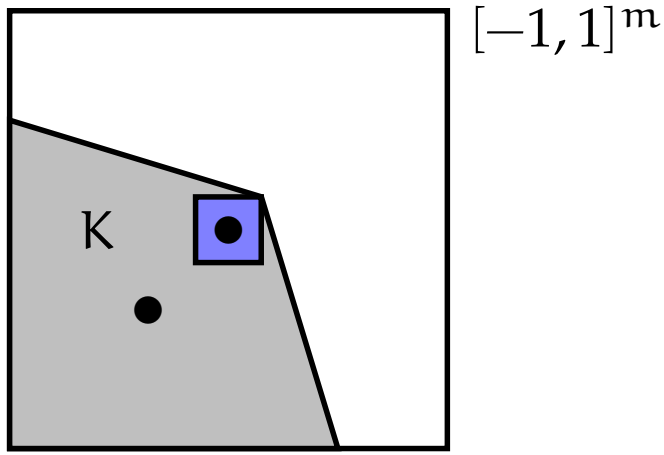


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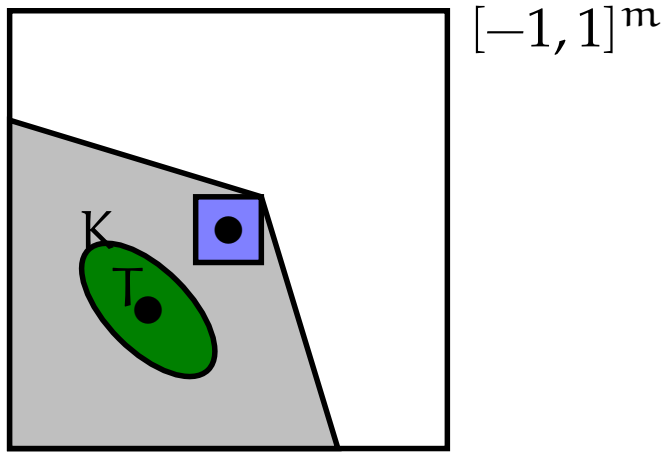


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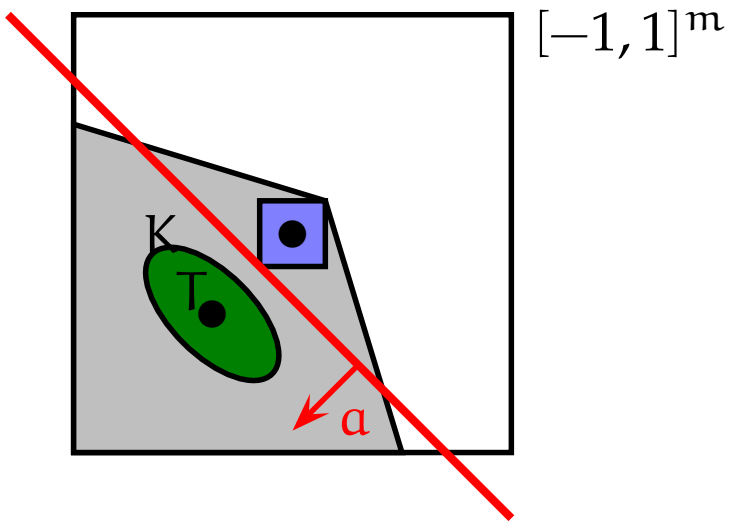
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- By hyperplane separation there exists $a \in \mathbb{R}^m$ so that

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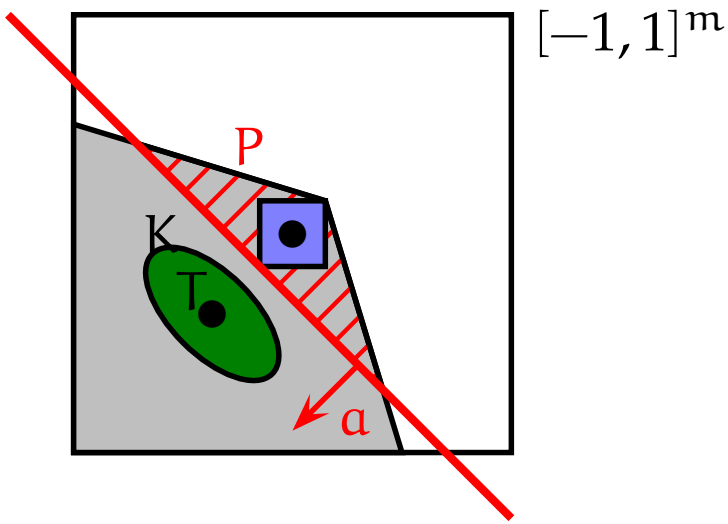
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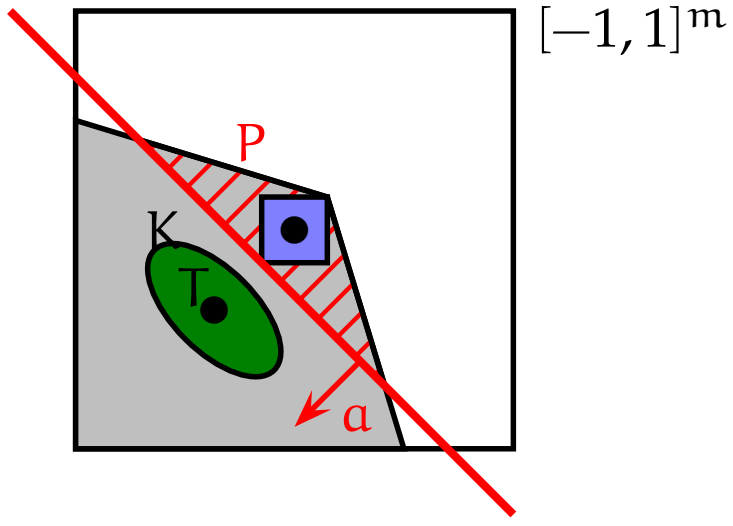
- Let $P := \{x \in K \mid \langle a, x \rangle \leqslant \varepsilon \|a\|_1\}$.



Size of the symmetrizer (4)

Claim

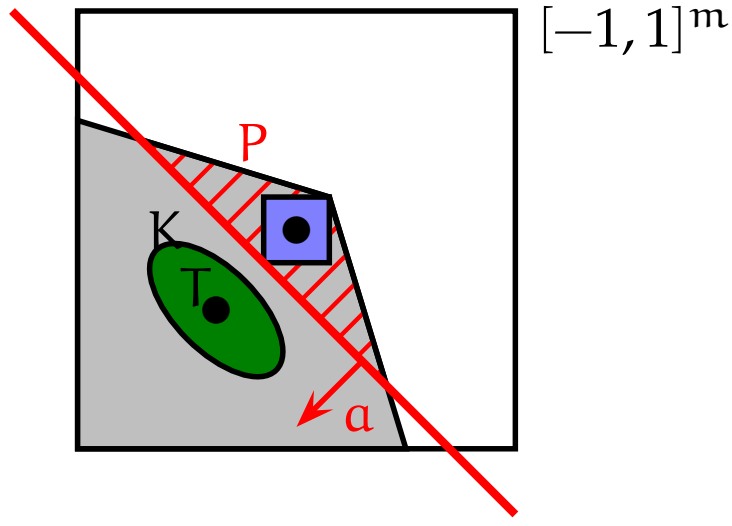
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Size of the symmetrizer (4)

Claim

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- With the claim,

$$\begin{aligned} f_K(\mathrm{bary}(P)) &\overset{P\subseteq K}{\geqslant} f_P(\mathrm{bary}(P)) \\ &\geqslant 2^{-m}\mathrm{Vol}_m(P) \\ &\geqslant 2^{-4m} \end{aligned}$$

- Then $\mathrm{bary}(P) \in T \to$ contradiction!

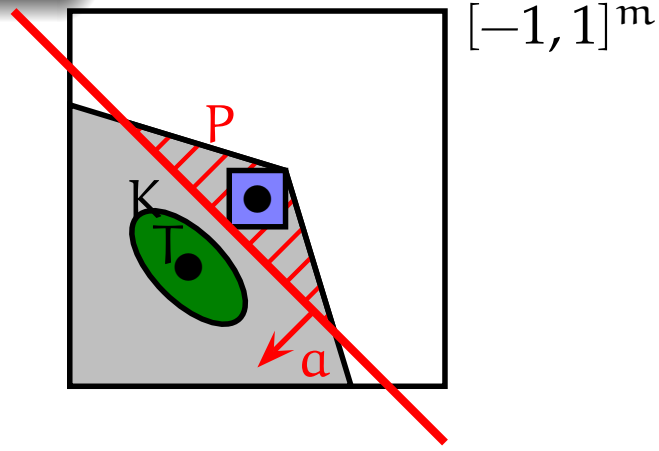
Size of the symmetrizer (5)

Claim

$\mathrm{Vol}_m(P) = \mathrm{Vol}_m(\{x \in K \mid \langle a, x\rangle \leqslant \varepsilon \|a\|_1\}) \geqslant 2^{-3m}$.

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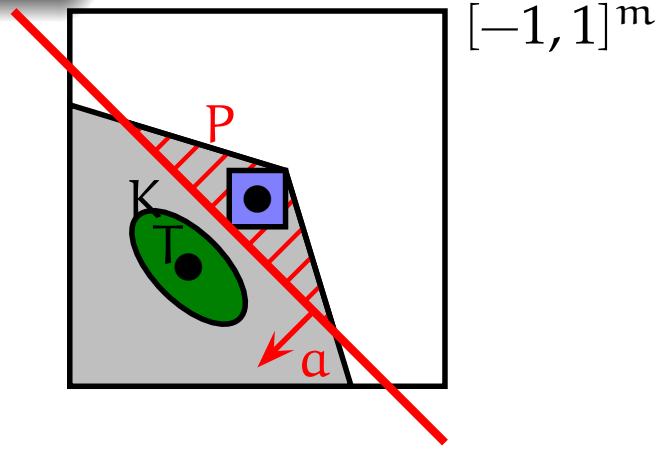
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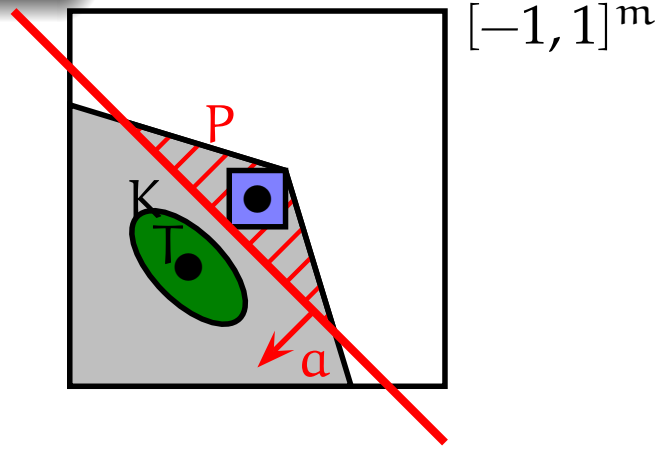
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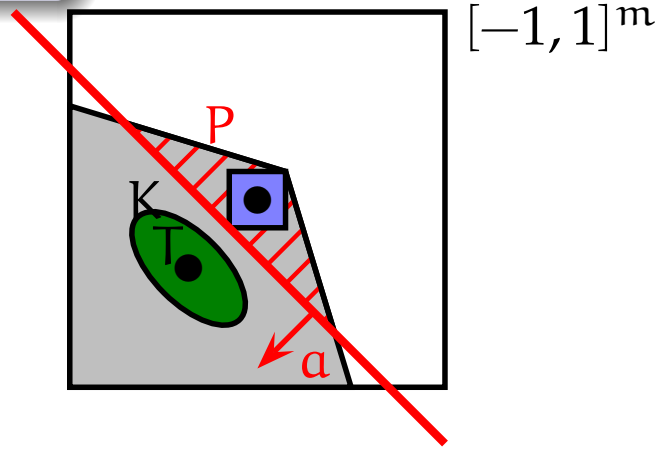
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- Then

$$Q := \left(\sqrt{\frac{|S|}{m}}\cdot\left\{x \in K_S \mid \langle a, x\rangle \leqslant \sqrt{\frac{m}{|S|}}\cdot\varepsilon\right\}\right) \times K_{[m]\setminus S} \subseteq P$$

- has

$\mathrm{Vol}_m(Q) \geqslant \left(\sqrt{\frac{|S|}{m}}\right)^{|S|}\cdot(\frac{1}{2}2^{-\varepsilon^2 m})\cdot 2^{|S|}\cdot(2^{-\varepsilon^2 m})\cdot 2^{m-|S|} \geqslant 2^{-3m}$. □

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Theorem

Let $K \subseteq [-1,1]^m$ be convex with $\mathrm{Vol}_S(K_S) \geqslant 2^{-\varepsilon^2 m} \cdot 2^{|S|}$ for all $S \subseteq [m]$ and $[-\varepsilon,\varepsilon]^m \subseteq K$. Then

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- It suffices to lower bound

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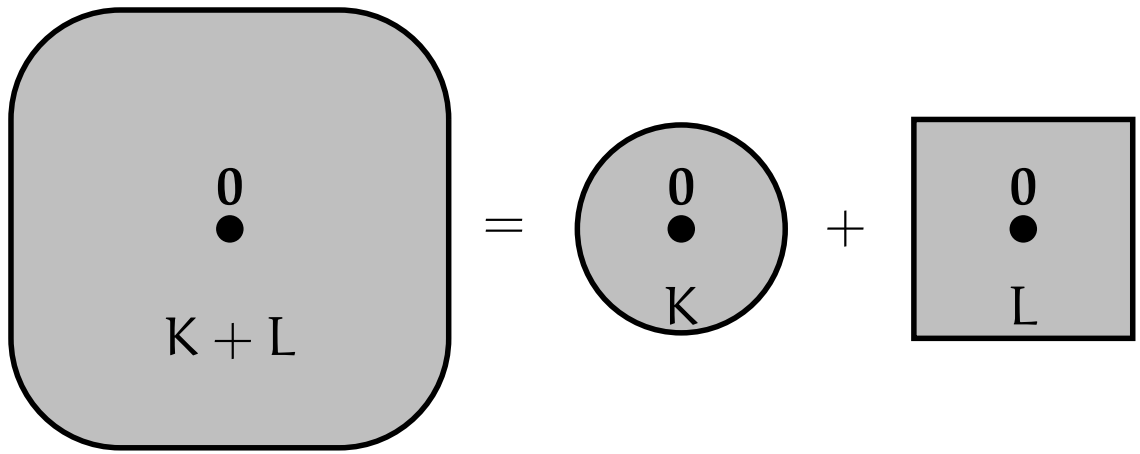
$$\Pr_{x \in \{-1,1\}^m} \left[\sum_{i=1}^m x_i \cdot |\langle a_i, y\rangle| \leqslant 0 \; \forall\, y \in \mathbb{R}^n \right]$$

- Idea: find a function f so this is "$f(x) > 0$" and use Boolean analysis.

Minkowski subtraction

- For convex sets $K, L \subseteq \mathbb{R}^n$, the **Minkowski sum** is

$$K + L = \{x + y \mid x \in K, y \in L\}.$$



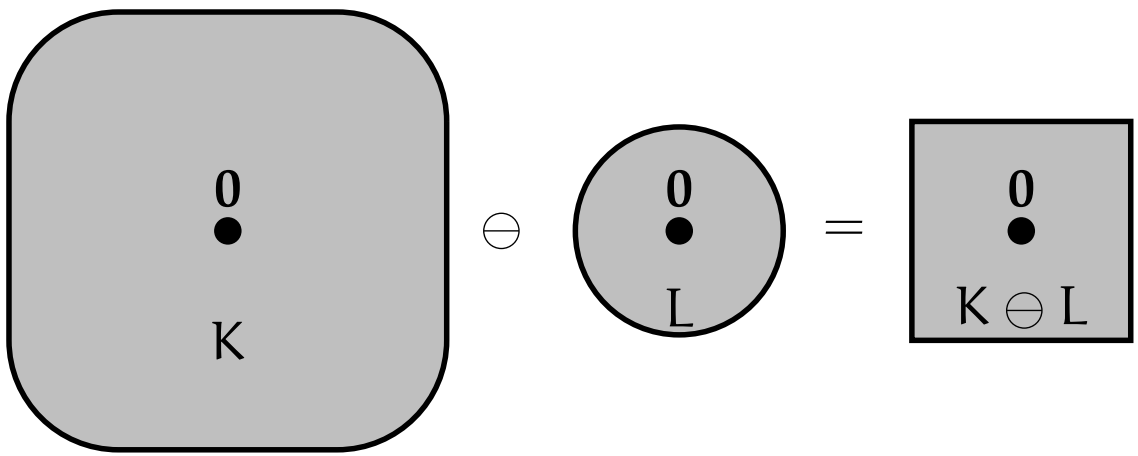
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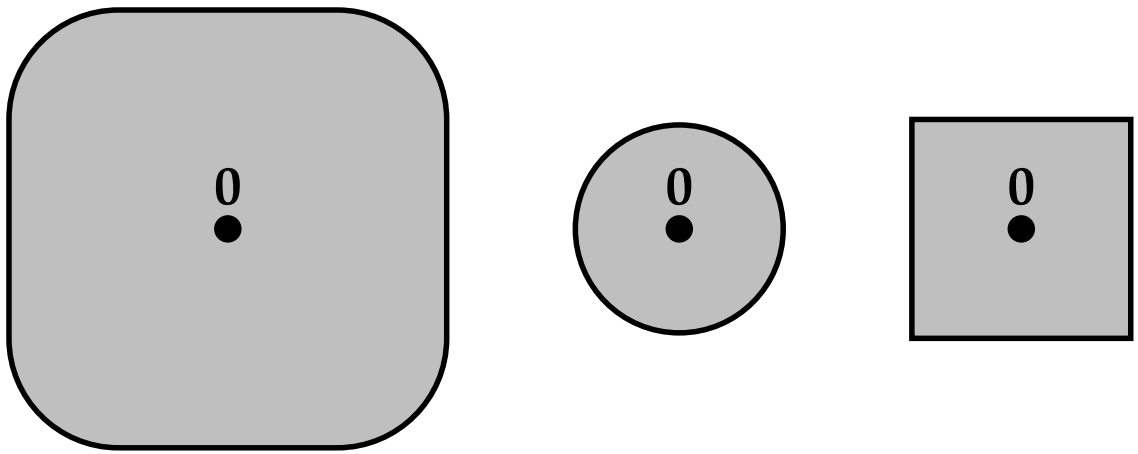
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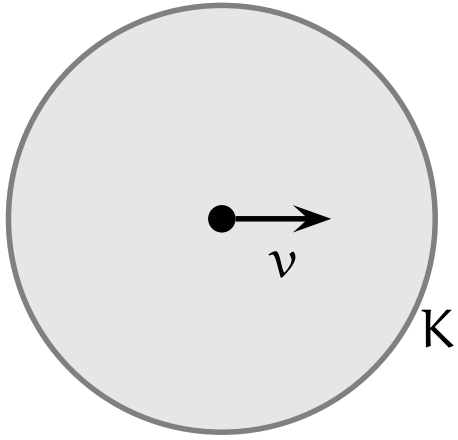
- Useful because: $K \ominus L \neq \emptyset \iff L \subseteq K$ for symmetric K, L.



Convexity of volume decrease

Lemma

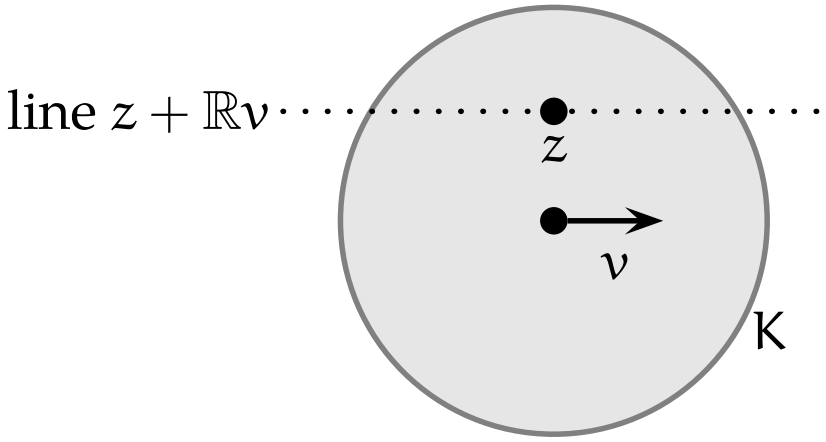
Let $K \subseteq \mathbb{R}^n$ be convex and let $v \in \mathbb{R}^n$. The function $f(s) := \mathrm{Vol}_n(K \ominus s[-v, v])$ is convex for $s \in [0, \infty)$.



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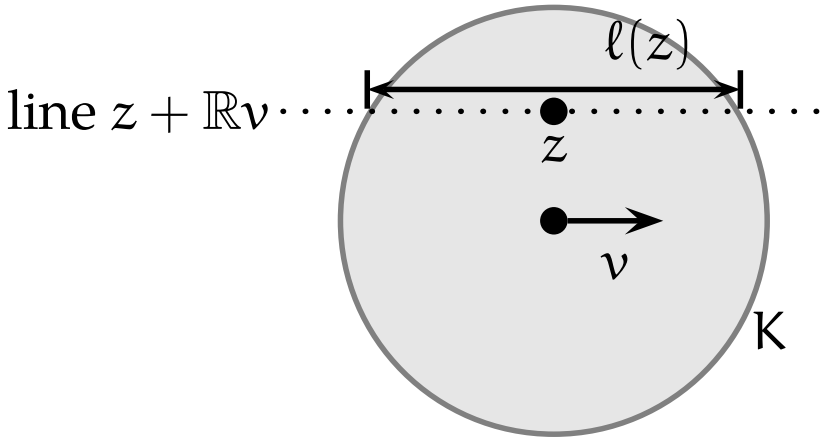


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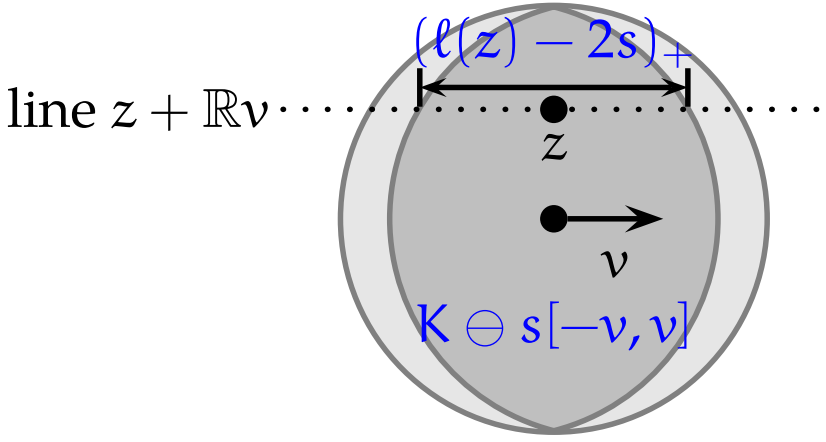


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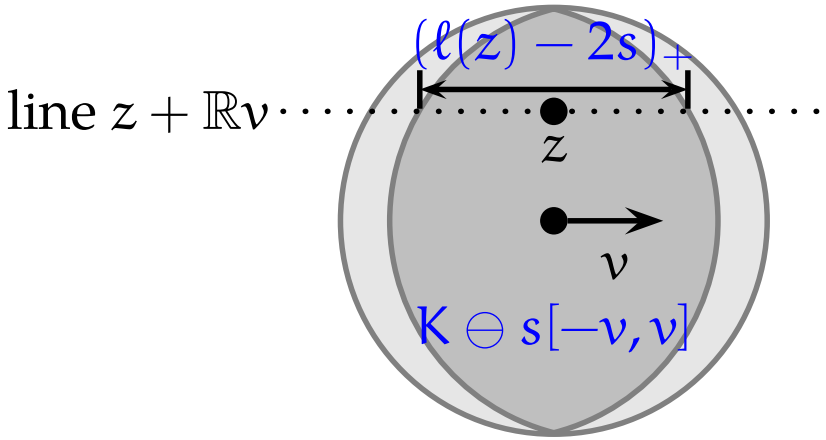


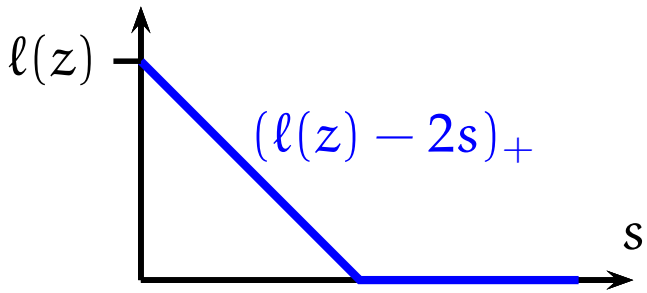
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- Then

$$\mathrm{Vol}_n(K \ominus s[-v, v]) = \int_{v^\perp} \underbrace{(\ell(z) - 2s\|v\|_2)_+}_{\text{convex in } s} \, dz$$

is an average of convex functions.

Inclusion of random zonotopes

- Consider zonotope $Z = \{\sum_{i=1}^{m} y_i A_i : |y_i| \leqslant 1\}$
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Lemma

Any f satisfying (i) and (ii) has $\Pr_{x \sim \{-1,1\}^m}[f(x) > 0] \geqslant e^{-2n}$.

- Uses Bonami's 1970 **hypercontractivity** theorem.

Finishing the argument

- It remains to note that $Z_{1+x} \subseteq Z$ is equivalent to

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Reproving [BSS'08]

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For PSD matrices $A_1, \ldots, A_m \in \mathbb{R}^{n \times n}$ with $\sum_{i=1}^{m} A_i = B$ and $0 < \varepsilon \leqslant \frac{1}{2}$ there is $w \in \mathbb{R}^m_{\geqslant 0}$ with $|\mathrm{supp}(w)| \leqslant O\left(\frac{n}{\varepsilon^2}\right)$ so that

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Other results + Follow-up

- Can reprove [BSS'08] (+ poly-time algo)
- Optimal Sparsifiers for Abelian Caley graphs
  [Basu, Kothari, Meka, Tudose '26]
- For sum $C := C_1 + \cdots + C_m \subseteq \mathbb{R}^n$ of symmetric convex sets, there is $s \geqslant \mathbf{0}$ with $|\mathrm{supp}(s)| \leqslant O(\frac{n}{\varepsilon^2})$ so that

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Thanks for your attention

Fourier analysis

Lemma

Let $f : [-1,1]^m \to \mathbb{R}_{\geqslant 0}$ be (i) separately convex, (ii) $f(\rho x) = \rho^n f(x)$ for all $x \in \{-1,1\}^m$ and $0 \leqslant \rho \leqslant 1$ and (iii) $f \neq 0$. Then $\Pr_{x \sim \{-1,1\}^m}[f(x) > 0] \geqslant c^n$.

- Let $\alpha := \Pr_{x \sim \{-1,1\}^m}[f(x) > 0]$.
- Pick $\rho \in (0,1)$ and $p \in (1,2)$ with $\rho = \sqrt{p-1}$.
- For any $x \in \{-1,1\}^m$,

$$T_\rho f(x) = \mathop{\mathbb{E}}_{z \sim N_\rho(x)}[f(z)] \overset{(i)}{\geqslant} f\Big(\mathop{\mathbb{E}}_{z \sim N_\rho(x)}[z]\Big) = f(\rho x) \overset{(ii)}{=} \rho^n f(x) \quad (*)$$

- Then

$$\rho^n \|f\|_2 \overset{(*)}{\leqslant} \|T_\rho f\|_2 \overset{(**)}{\leqslant} \|f\|_p \overset{(***)}{\leqslant} \|f\|_2 \cdot \alpha^{\frac{1}{p}-\frac{1}{2}}$$

  $(**)$ = hypercontractivity, $(***)$ =Hölder.
- Rearrange for α. □